

CharDiff-LP: A Diffusion Model with Character-Level Guidance for License Plate Image Restoration

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1. Motivation

- Without textual guidance, diffusion models invent plausible but wrong characters under severe degradation (e.g., 7→3, M→N).
- String-level priors help, but global cross-attention lets guidance spill across neighboring characters.

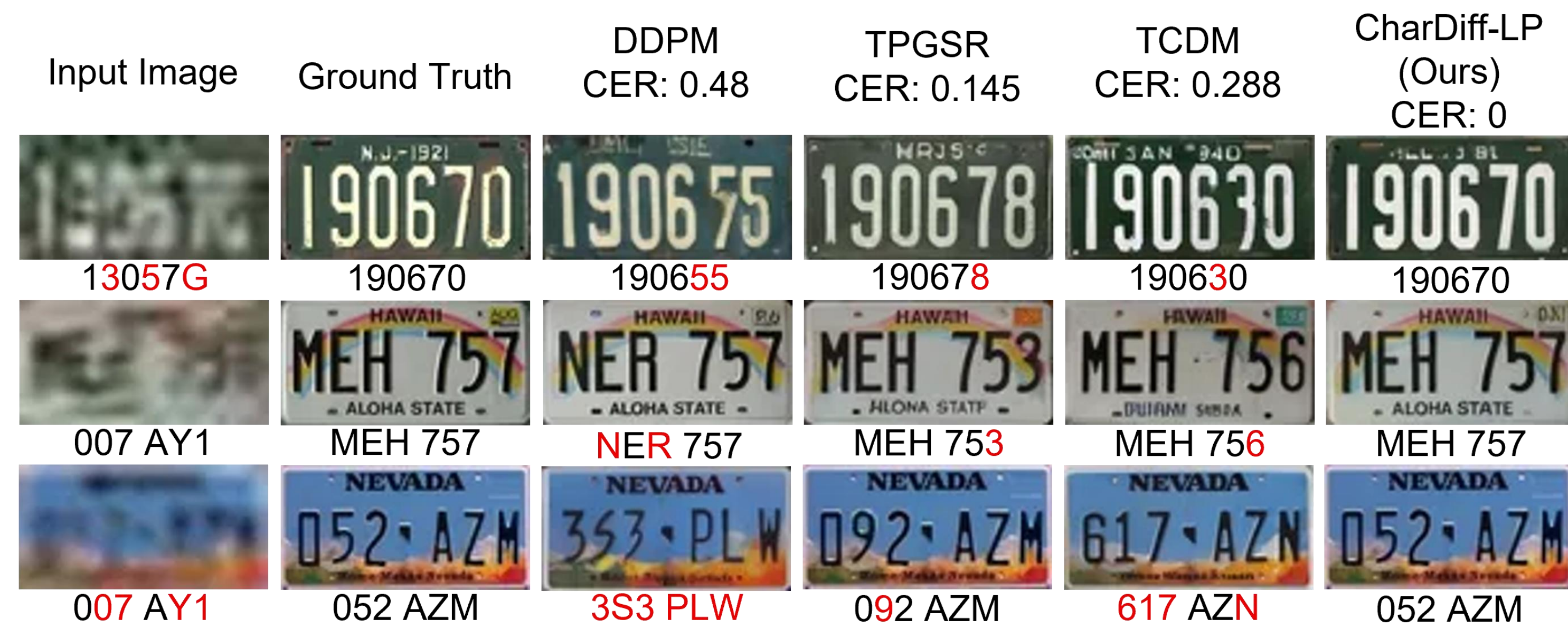


Fig. 1. Baselines distort or change character identities under severe degradation; CharDiff-LP restores them correctly.

3. Framework & CHARM Module

A CRAFT-based segmentator and a degradation-aware OCR model (MGP-STR) extract character regions and identities, which are converted into embeddings e_i and spatial masks M_i for CHARM.

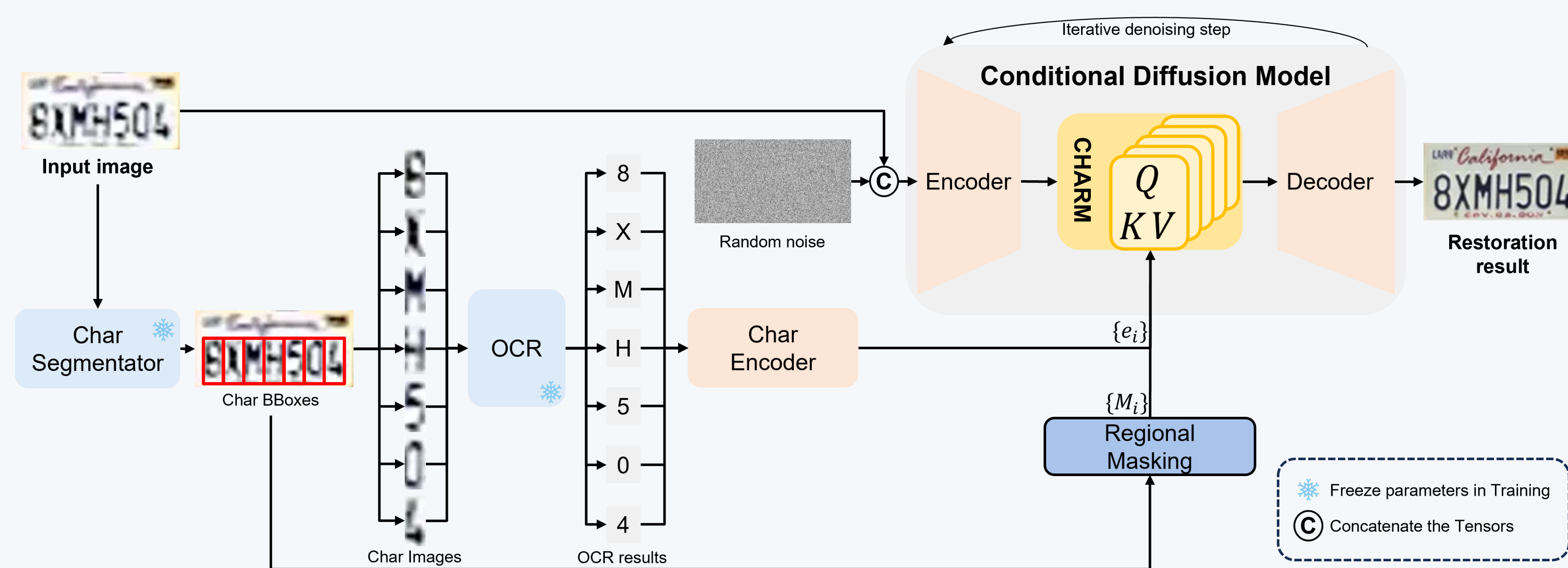


Fig. 2. Overview: character priors (embeddings + masks) guide the conditional diffusion model via CHARM throughout denoising.

4. Qualitative & Ablation



Fig. 5. CharDiff-LP preserves plate identities where baselines distort characters (Dashcam-LP).

Variant (Roboflow-LP)	CER ↓	LPR Acc. ↑
DDPM (no text prior)	0.2118	0.514
+ String-level prior	0.1539	0.575
+ Character-level prior	0.1187	0.621
+ CHARM (full, Ours)	0.0810	0.697

Spatial confinement alone adds +7.6pp plate-level accuracy (0.621 → 0.697).

2. Main Results

28.3%

relative CER reduction vs. the best baseline (TCDM fine-tuned) on Roboflow-LP: 11.3% → 8.1%

Roboflow-LP (paired real HQ / synthetic LQ)

Method	PSNR ↑	LPIPS ↓	CER ↓	LPR Acc. ↑
DDPM (no text prior)	23.98	0.5022	0.212	0.514
TPGSR (fine-tuned)	23.75	0.4340	0.157	0.637
TCDM (fine-tuned)	24.16	0.4114	0.113	0.667
CharDiff-LP (Ours)	24.35	0.3832	0.081	0.697

Dashcam-LP (unpaired real LQ)

Method	CER ↓	LPR Acc. ↑
DDPM (no text prior)	0.2118	0.5458
TPGSR (fine-tuned)	0.1714	0.6266
TCDM (fine-tuned)	0.1613	0.6367
CharDiff-LP (Ours)	0.1512	0.6965

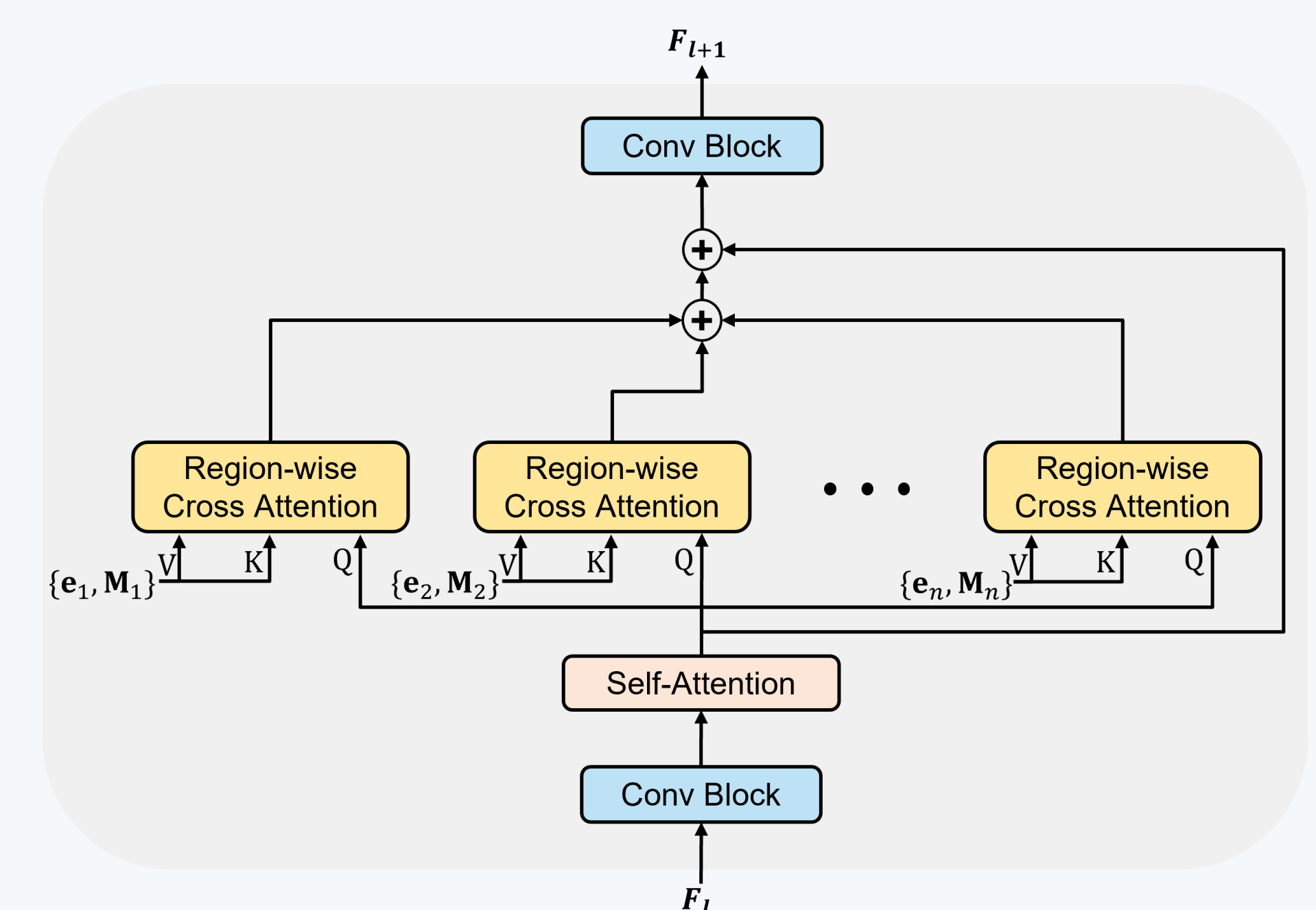


Fig. 3. CHARM: region-wise masked cross-attention in the U-Net middle block.

5. CHARM & Attention Visualization

- CHARM (Character-guided Attention through Region-wise Masking): embedding e_i attends only inside its spatial mask M_i .
- The $(1 - M_i) \cdot (-\infty)$ term zeroes attention scores outside the region.

$$A_i = \text{softmax}\left(\frac{Q^T K_i}{\sqrt{d}} + (1 - M_i) \cdot (-\infty)\right)$$
$$F_{l+1} = F_l + \sum_i \text{Reshape}(V_i A_i^T)$$

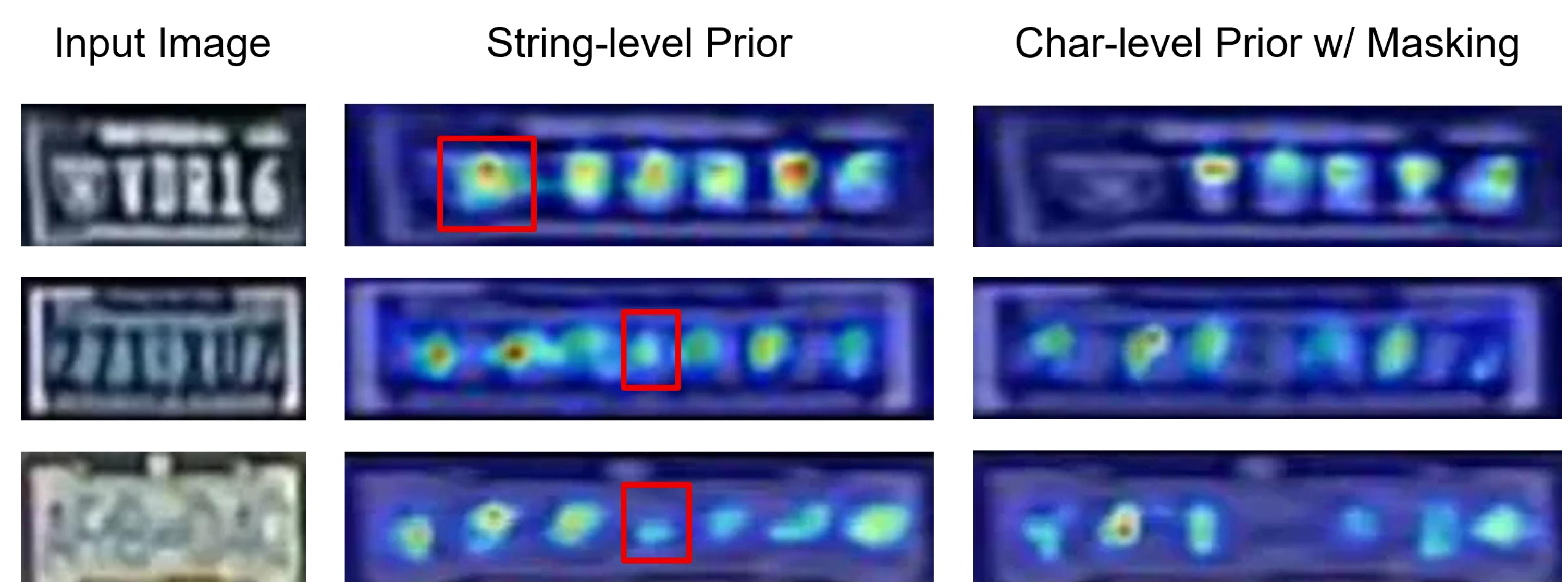


Fig. 6. String-level attention leaks into non-character regions (red boxes); CHARM confines each prior to its character mask.

Fine-grained character-level guidance outperforms string-level guidance, but identity alone is not enough: gains are maximized when each character's guidance is confined to its correct spatial region.

Takeaways

Contribution

To our knowledge, the first diffusion-based LP restoration model with character-level guidance via spatially masked cross-attention (CHARM).

Limitation

Relies on an external segmentator and OCR; severe recognition errors can propagate to restoration.

Future work

Joint training of recognition and restoration; uncertainty modeling.

Paper link

